

# Souha Alioua

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## PROFILE SUMMARY

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Data Analyst with hands-on experience building automated Python data pipelines, querying large-scale datasets in SQL Server, and developing Tableau dashboards for public-sector decision-making. Strong background in statistical analysis and data validation, with a focus on delivering reliable, actionable insights from complex datasets.

## EDUCATION

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### Dickinson College

Carlisle, PA

*B.S. Mathematics*

*Jan 2021 – Dec 2024*

- **Relevant Coursework:** Principles of Object-Oriented Design, Computational Mathematics, Tools and Techniques for Software Development, Computing Abstractions, Large-scale/Open-Source Software Development, Intro to Data Science, Data Systems for Data Analytics
- **Awards:** Henry P. Cannon Memorial Prize in Mathematics (2022), John Dickinson Scholarship, Dean's List

## TECHNICAL SKILLS, CERTIFICATIONS & ACHIEVEMENTS

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**Tools & Software:** Microsoft Office (Excel, PPT, Outlook), Python (Pandas, Matplotlib, Seaborn), Tableau, SQL (MSSQL Server, MySQL), HTML/CSS, JavaScript, Power BI, IBM Cognos Analytics, Looker Studio, MS Access, ArcGIS

**Certifications:** IBM Data Analytics Professional Certificate, Machine Learning Specialization (expected May 2026)

**Languages:** English, French, Arabic

## WORK EXPERIENCE

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### Maryland Department of Planning – State Data & Geospatial Analysis Center

Baltimore, MD

*Data Analyst*

*Feb 2025 – Present*

- Automated 10+ Python/API data pipelines for demographic, economic, and health datasets, reducing manual processing time by ~10 hours/week and improving reporting reliability
- Developed Selenium-based web scraping workflows to collect hard-to-access public data, expanding datasets by 1,000+ records
- Built and validated large-scale datasets in SQL Server, implementing data quality checks and version control (GitLab) to improve accuracy and reproducibility
- Applied statistical methods to generate municipal-level population projections, supporting long-term planning and policy decisions

## PROJECTS

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### Personal Project - HR Analytics Dashboard | [Link](#)

*Jan 2025*

*Data Analyst | HR Analytics & Attrition Insights Dashboard*

- Processed and analyzed a 1,400+ employee dataset using Python (Pandas), creating structured features for KPI tracking and workforce segmentation
- Designed an interactive Tableau dashboard with filters and drilldowns to analyze attrition trends by department, age, gender, and education
- Delivered insights on key drivers of employee attrition to support HR decision-making

### Dickinson College Capstone Project - BookScape: Book Ratings & Trends Analysis Tool | [Link](#)

*Dec 2024*

*Data Analyst / Developer | Goodreads: Web Scraping, Data Cleaning, Trend Analysis*

- Scraped 1,500+ book records using Python, handling dynamic content and pagination
- Cleaned and standardized raw data into analysis-ready datasets
- Built Power BI visualizations to identify trends in ratings and reviews across genres